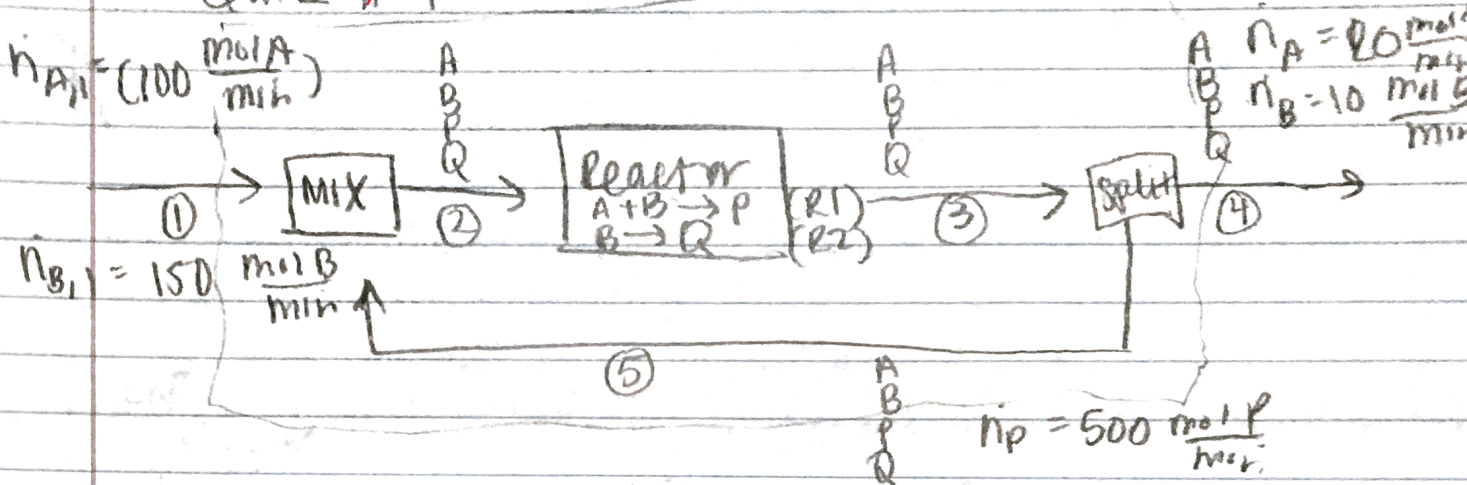


Sweeney

ChE 210

Quiz #4

(4)



Material balance:

A:

Reactor: $\dot{n}_{A,3} = (1 - f_{C,A}) \dot{n}_{A,2}$

Splitter: $\dot{n}_{A,4} + \dot{n}_{A,5} = \dot{n}_{A,3}$

Mixer: $\dot{n}_{A,1} + \dot{n}_{A,5} = \dot{n}_{A,2}$

Mixer: Species balances

A:

$100 = \dot{n}_{2,A} + \dot{n}_{5,A}$

B:

$150 = \dot{n}_{2,B} + \dot{n}_{5,B}$

↑?

↑

5 goes into mixer

Reactor:

A:

$\dot{n}_{A,3} = \dot{n}_{A,2} - \xi_1$

B:

$\dot{n}_{B,3} = \dot{n}_{B,2} - \xi_1 - \xi_2$

P:

$\dot{n}_{P,3} = \dot{n}_{P,2} + \xi_1$

Q:

$\dot{n}_{Q,3} = \dot{n}_{Q,2} + \xi_2$

Splitter

A:

$\dot{n}_{A,3} = \dot{n}_{A,4} + \dot{n}_{A,5} = 20 + \dot{n}_{A,5}$

B:

$\dot{n}_{B,3} = \dot{n}_{B,4} + \dot{n}_{B,5} = 10 + \dot{n}_{B,5}$

P:

$\dot{n}_{P,3} = \dot{n}_{P,4} + \dot{n}_{P,5} = \dot{n}_{P,4} + 500$

Q:

$\dot{n}_{Q,3} = \dot{n}_{Q,4} + \dot{n}_{Q,5}$

Overall:

$\dot{n}_{A,1} + \dot{n}_{B,1} = \dot{n}_{A,4} + \dot{n}_{B,4} + \dot{n}_{P,4} + \dot{n}_{Q,4}$

$250 = 20 + 10 + \dot{n}_{P,4} + \dot{n}_{Q,4}$

$220 = \dot{n}_{P,4} + \dot{n}_{Q,4}$

why add all species together?